MATH 152 **EXAM IIA** NAME Solutions

Mrs. Bonny Tighe 50 points

7.7-11.7 TA_____ Wed 4/3/2024

PART II

1. (10 points) Complete the following tests

GEOMETRIC SERIES $\sum_{n=0}^{\infty} a_n (r)^n$ is Convergent if $|r| < 1$

THE COMPARISON TEST Suppose $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with positive terms.

i) If $\sum_{n=1}^{\infty} b_n$ is convergent and $a_n \leq b_n$ for all n, then $\sum_{n=1}^{\infty} a_n$ is also convergent

ii) If $\sum_{n=1}^{\infty} b_n$ is divergent and $a_n \geq b_n$ for all n, then $\sum_{n=1}^{\infty} a_n$ is also divergent

THE LIMIT COMPARISON TEST

Suppose that $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with positive terms . If $\lim_{n\to\infty} \frac{a_n}{b_n} = c$ where c is positive + finite, then either both series converge or both diverge.

Name the test: Test for Divergence If $\lim_{n\to\infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ is divergent

THE INTEGRAL TEST Suppose f is a positive, continuous and decreasing function on $[1, \infty)$ and let $a_n = f(n)$ then the series $\sum_{n=1}^{\infty} a_n$ is convergent if and only if $\int_a^{\infty} f(n)\,dn$ is convergent.

P-SERIES $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is divergent if $p \leq 1$

RATIO TEST

If $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L$

$L < 1$ then $\sum a_n$ is absolutely convergent

$L > 1$ then $\sum a_n$ is divergent

$L = 1$ then the test is inconclusive

ROOT TEST

If $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L$ and

$L < 1$ then $\sum a_n$ is absolutely convergent

$L > 1$ then $\sum a_n$ is divergent

$L = 1$ then the root test is inconclusive

THE ALTERNATING SERIES TEST

If i) if $b_{n+1} \leq b_n$ for all n and ii) $\lim_{n\to\infty} b_n = 0$ Then $\sum_{n=1}^{\infty} (-1)^{n-1} b_n$ is Convergent

REMAINDER ESTIMATE FOR THE INTEGRAL TEST

Suppose $a_n = f(n)$ where f is a continuous, positive, decreasing function for $x \geq n$, and $\sum a_n$ is convergent . If $R_n = S - S_n$, then $\int_{n+1}^{\infty} f(n)\,dn \leq R_n \leq \int_n^{\infty} f(n)\,dn$

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2. (~~10~~ points). Use the **Integral test** to determine if the series is divergent or convergent.

$$\sum_{n=1}^{\infty} \frac{e^{1/n}}{n^2}$$

$u = \frac{1}{n}$ $du = -\frac{1}{n^2}dn$

$$\lim_{t\to\infty} \int_1^t e^{1/n}\left(\frac{1}{n^2}\right)dn = \lim_{t\to\infty} -e^{1/n}\Big]_1^t$$

$$\lim_{t\to\infty} \left[-e^{1/t} - \left(-e^{1/1}\right)\right] = -1+e$$ which is finite

$\sum_{n=1}^{\infty} \frac{e^{1/n}}{n^2}$ is also convergent

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3. (~~10~~ points) Find the length of the arc of the curve $y = \frac{x^3}{6} + \frac{1}{2x}$, $\frac{1}{2} \le x \le 1$

$y = \frac{1}{6}x^3 + \frac{1}{2}x^{-1}$

$dy/dx = \frac{1}{2}x^2 - \frac{1}{2}x^{-2}$

$= \frac{1}{2}x^2 - \frac{1}{2x^2}$

$(dy/dx)^2 = \frac{1}{4}x^4 - \frac{1}{2} + \frac{1}{4x^4}$

$$L = \int_{1/2}^{1} \sqrt{1 + (dy/dx)^2}\, dx$$

$$= \int_{1/2}^{1} \sqrt{\frac{1}{4}x^4 + \frac{1}{2} + \frac{1}{4x^4}}\, dx$$ "one of those"

$$\int_{1/2}^{1} \sqrt{\left(\frac{1}{2}x^2 + \frac{1}{2x^2}\right)^2}\, dx = \int_{1/2}^{1} \left(\frac{1}{2}x^2 + \frac{1}{2}x^{-2}\right) dx$$

$$\frac{1}{2}\cdot\frac{1}{3}x^3 + \frac{1}{2}\cdot\frac{1}{-1}x^{-1}\Big]_{1/2}^{1} = \frac{1}{6}x^3 - \frac{1}{2x}\Big]_{1/2}^{1} =$$

$$\left(\frac{1}{6} - \frac{1}{2}\right) - \left(\frac{1}{6}\left(\frac{1}{8}\right) - \frac{1}{1}\right) = \frac{1}{6} - \frac{1}{2} - \frac{1}{48} + 1$$

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4. (10 points) Determine if the following improper integral is convergent or divergent. If convergent, evaluate.

discontinuous $\int_1^3 \frac{x}{\sqrt{x^2-1}}\,dx$

$\lim_{t\to 1^+} \int_t^3 (x^2-1)^{-1/2}\, x\,dx =$ $\quad u = x^2-1$, $du = 2x\,dx$

$\lim_{t\to 1^+} \frac{1}{2}\cdot\frac{1}{1/2}(x^2-1)^{1/2}\Big]_t^3 = \lim_{t\to 1^+} \sqrt{x^2-1}\Big]_t^3 =$

$\lim_{t\to 1^+}\left(\sqrt{9-1} - \sqrt{t^2-1}\right) = \sqrt{8} - 0 = \sqrt{8}$ a finite limit

So $\int_1^3 \frac{x}{\sqrt{x^2-1}}\,dx$ is convergent.

5. (10 points) Determine whether the series is convergent or divergent and if it is convergent **find the sum.**

a) $\sum_{n=1}^{\infty} \frac{1}{n^2+3n}$

$\frac{A}{n} + \frac{B}{n+3} = \frac{1}{n(n+3)}$

$A(n+3) + Bn = 1$

$n=0 \quad A = 1/3$

$n=-3 \quad B = -1/3$

$\sum_{n=1}^{\infty}\left(\frac{1/3}{n} - \frac{1/3}{n+3}\right) =$

$\frac{1/3}{1} + \frac{1/3}{2} + \frac{1/3}{3} + \cancel{\frac{1/3}{4}} + \cancel{\frac{1/3}{5}} + \cancel{\frac{1/3}{6}}$

$\cancel{-\frac{1/3}{4}} - \cancel{\frac{1/3}{5}} - \cancel{\frac{1/3}{6}}$

telescopic

The sum of the series is

$\frac{1}{3}+\frac{1}{6}+\frac{1}{9}$ because I found the sum, the series $\sum_{n=1}^{\infty}\frac{1}{n^2+3n}$ is Convergent

b) $\sum_{n=1}^{\infty} \frac{(-4)^n}{e^{2n+2}}$ $\quad \frac{(-4)(-4)^{n-1}}{(e^2)^{n+1}}$

$\sum_{n=1}^{\infty} \frac{(-4)(-4)^{n-1}}{e^4(e^2)^{n-1}}$ $\quad r = \left|\frac{-4}{e^2}\right| < 1$

Convergent by the Geometric series test so

Sum $= \frac{a}{1-r}$

$S = \frac{-4/e^4}{1-(-4/e^2)} =$

$\frac{-4}{e^4+4e^2}$

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6. (10 points) Test the following series for absolute convergence, conditional convergence or divergence. State the test you are using and show all your work.

$b_n = \frac{1}{2n+3}$

3 a) $\sum_{n=1}^{\infty} \frac{n+1}{\sqrt{n^5+3n^2+1}}$

Compare to $\sum_{n=1}^{\infty} \frac{1}{n^{3/2}}$ a convergent p-series $p = 3/2 > 1$

Limit Comparison Test

$\lim_{n\to\infty} \left(\frac{n+1}{\sqrt{n^5+3n^2+1}} \cdot \frac{n^{3/2}}{1} \right) =$

$\lim_{n\to\infty} \left(\frac{n^{7/2}+n^{3/2}}{\sqrt{n^5+3n^2+1}} \right) = 1 > 0$ and finite

$\therefore \sum_{n=1}^{\infty} \frac{n+1}{\sqrt{n^5+3n^2+1}}$ is Absolutely Convergent

4 b) $\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+3}$

- Compare to $\sum_{n=1}^{\infty} \frac{1}{n}$ a divergent p-series $p = 1 \leq 1$
- Limit Comparison Test:

$\lim_{n\to\infty} \left(\frac{1}{n} \cdot \frac{2n+3}{1} \right) = 2 > 0$ and finite so both are divergent.

- next use the Alternating Series Test

(i) $\frac{1}{2(n+1)+3} \leq \frac{1}{2n+3}$

(ii) $\lim_{n\to\infty} \left(\frac{1}{2n+3} \right) = 0$

$\therefore \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+3}$ is Conditionally Convergent

3 c) $\sum_{n=1}^{\infty} \frac{4 \cdot 7 \cdot 10 \cdots (3n+1)}{4^n n!}$

Ratio Test

$\lim_{n\to\infty} \left| \frac{(4 \cdot 7 \cdot 10 \cdots (3n+1)(3n+4))}{4 \cdot 4^n (n+1) n!} \cdot \frac{4^n n!}{(4 \cdot 7 \cdot 10 \cdots (3n+1))} \right|$

$\lim_{n\to\infty} \left| \frac{3n+4}{4(n+1)} \right| = 3/4 < 1$ so absolutely convergent

5 $\qquad r = y$

7. (~~10~~ points) The given curve is rotated about the x-axis. Find the area of the resulting surface. $S = 2\pi \int_a^b radius\, arclength$ $\qquad y^{2/3} + x^{2/3} = 1,\ 0 \le x \le 1$

dx

$$S = 2\pi \int_0^1 y \sqrt{1 + (dy/dx)^2}\, dx$$

$$y = (1 - x^{2/3})^{3/2}$$

$$dy/dx = \frac{3}{2}(1 - x^{2/3})^{1/2}\left(-\frac{2}{3}x^{-1/3}\right)$$

$$= \frac{-\sqrt{1 - x^{2/3}}}{x^{1/3}}$$

$$(dy/dx)^2 = \frac{1 - x^{2/3}}{x^{2/3}}$$

$$S = 2\pi \int_0^1 (1 - x^{2/3})^{3/2} \sqrt{1 + \frac{1 - x^{2/3}}{x^{2/3}}}\, dx$$

$$S = 2\pi \int_0^1 (1 - x^{2/3})^{3/2} \sqrt{\frac{x^{2/3} + 1 - x^{2/3}}{x^{2/3}}}\, dx$$

$$2\pi \int_0^1 (1 - x^{2/3})^{3/2} \sqrt{\frac{1}{x^{2/3}}}\, dx = 2\pi \int_0^1 (1 - x^{2/3})^{3/2} \left(\frac{1}{x^{1/3}}\right) dx$$

$$u = 1 - x^{2/3}$$

$$du = -2/3\, x^{-1/3}\, dx$$

$$= dx$$

$$-\frac{3}{2} x^{1/3}\, du$$

$$2\pi \int -\frac{3}{2} u^{3/2}\, du$$

$$2\pi \left[-3/2 \cdot \frac{1}{5/2} u^{5/2} \right]$$

$$2\pi \left[\left(-\frac{3}{2} \cdot \frac{2}{5}\right)(1 - x^{2/3})^{5/2} \right]_0^1$$

$$2\pi \left(-\frac{3}{5}\right)\left[0^{5/2} - 1^{5/2}\right] = -\frac{6}{5}\pi[-1] = 6\pi/5$$

MATH 152
Mrs. Bonny Tighe

EXAM II

50 points
7.7-11.7
PART II

NAME Solutions

TA_____ Wed 4/3/2024

1. (10 points) Complete the following tests

GEOMETRIC SERIES $\sum_{n=0}^{\infty} a_n (r)^n$ is divergent if $|r| \geq 1$

THE COMPARISON TEST Suppose $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with positive terms.

i) If $\sum_{n=1}^{\infty} b_n$ is convergent and $a_n \leq b_n$ for all n, then $\sum_{n=1}^{\infty} a_n$ is also convergent

ii) If $\sum_{n=1}^{\infty} b_n$ is divergent and $a_n \geq b_n$ for all n, then $\sum_{n=1}^{\infty} a_n$ is also divergent

THE LIMIT COMPARISON TEST

Suppose that $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are series with positive terms. If $\lim_{n\to\infty} \frac{a_n}{b_n} = c$ where c is > 0 and finite, then either both series converge or both diverge.

Name the test: Test for Divergence If $\lim_{n\to\infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ is divergent

THE INTEGRAL TEST Suppose f is a positive, decreasing and continuous function on $[1, \infty)$ and let $a_n = f(n)$ then the series $\sum_{n=1}^{\infty} a_n$ is convergent if and only if $\int_a^{\infty} f(n)\,dn$ is convergent.

P-SERIES $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent if $p > 1$

RATIO TEST

If $\lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| = L$

$L < 1$ then $\sum a_n$ is absolutely convergent
$L > 1$ then $\sum a_n$ is divergent
$L = 1$ then the test is inconclusive

ROOT TEST

If $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L$ and

$L < 1$ then $\sum a_n$ is absolutely convergent
$L > 1$ then $\sum a_n$ is divergent
$L = 1$ then the root test is inconclusive

THE ALTERNATING SERIES TEST

If i) if $b_{n+1} \leq b_n$ for all n and ii) $\lim_{n\to\infty} b_n \neq 0$ Then $\sum_{n=1}^{\infty} (-1)^{n-1} b_n$ is Divergent

REMAINDER ESTIMATE FOR THE INTEGRAL TEST

Suppose $a_n = f(n)$ where f is a continuous, positive, decreasing function for $x \geq n$, and $\sum a_n$ is convergent. If $R_n = S - S_n$, then $\int_{n+1}^{\infty} f(n)\,dn \leq R_n \leq \int_n^{\infty} f(n)\,dn$

2. (10 points). Use the **Integral test** to determine if the series is divergent or convergent.

$$\sum_{n=1}^{\infty} \frac{e^n}{e^{2n}+1}$$

$f(n) = \frac{e^n}{e^{2n}+1}$ which is continuous, positive and decreasing for $n \geq 1$

$$\lim_{t\to\infty} \int_1^t \frac{e^n}{e^{2n}+1}\,dn = \lim_{t\to\infty} \tan^{-1}(e^n)\Big]_1^t$$

$$\lim_{t\to\infty} \left[\tan^{-1}(e^t) - \tan^{-1}(e^1)\right] = \pi/2 - \tan^{-1}(e)$$ which is finite so is convergent

$\sum_{n=1}^{\infty} \frac{e^n}{e^{2n}+1}$ is also convergent

3. (10 points) Find the length of the arc of the curve $x = \frac{y^4}{8} + \frac{1}{4y^2}$ $1 \leq y \leq 2$

$$L = \int_1^2 \sqrt{1 + (dx/dy)^2}\,dy$$

$$L = \int_1^2 \sqrt{\tfrac{1}{4}y^6 + \tfrac{1}{2} + \tfrac{1}{4y^6}}\,dy$$

$$L = \int_1^2 \sqrt{\left(\tfrac{1}{2}y^3 + \tfrac{1}{2y^3}\right)^2}\,dy$$

$$L = \int_1^2 \left(\tfrac{1}{2}y^3 + \tfrac{1}{2}y^{-3}\right) dy$$

$$L = \tfrac{1}{8}y^4 - \tfrac{1}{4}y^{-2}\Big]_1^2$$

$$\left(\tfrac{1}{8}(2^4) - \tfrac{1}{4(4)}\right) - \left(\tfrac{1}{8} - \tfrac{1}{4}\right)$$

$$2 - \tfrac{1}{16} - (-1/8)$$

$$2 - \tfrac{1}{16} + \tfrac{1}{8} = 2 + 1/16$$

$$x = \tfrac{1}{8}y^4 + \tfrac{1}{4}y^{-2}$$

$$dx/dy = \tfrac{1}{2}y^3 - \tfrac{1}{2}y^{-3}$$

$$(dx/dy)^2 = \left(\tfrac{1}{2}y^3 - \tfrac{1}{2y^3}\right)^2$$

$$= \tfrac{1}{4}y^6 - \tfrac{1}{2} + \tfrac{1}{4y^6}$$

"one of those"

4. (10 points) Determine if the following improper integral is convergent or divergent. If convergent, evaluate.

$$\int_2^3 \frac{5x}{\sqrt{x^2-4}}\,dx \qquad \lim_{t\to 2^+} \int_t^3 (x^2-4)^{-1/2}\,5x\,dx =$$

$$\lim_{t\to 2^+} \frac{5}{2}\cdot\frac{1}{1/2}(x^2-4)^{1/2}\Big]_t^3 =$$

$$\lim_{t\to 2^+} 5\sqrt{x^2-4}\Big]_t^3 = \lim_{t\to 2^+}\left[5\sqrt{9-4} - 5\sqrt{t^2-4}\right] =$$

$$\left[5\sqrt{5} - 5\sqrt{0}\right] = 5\sqrt{5}$$ so Convergent

5. (10 points) Determine whether the series is convergent or divergent and if it is convergent find the sum.

a) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2+2n}$

$$\frac{A}{n} + \frac{B}{n+2} = \frac{1}{n(n+2)}$$

$A(n+2) + Bn = 1$ telescoping

$n=0$ $\quad A = 1/2$

$n=-2$ $\quad B = -1/2$

$$\sum_{n=1}^{\infty}\left(\frac{1/2}{n} - \frac{1/2}{n+2}\right)$$

$$\frac{1}{2} + \frac{1/2}{2} + \cancel{\frac{1/2}{3}} + \cancel{\frac{1/2}{4}} + \cancel{\frac{1/2}{5}} + \cdots$$

$$-\cancel{\frac{1/2}{3}} - \cancel{\frac{1/2}{4}} - \cancel{\frac{1/2}{5}}$$

$\frac{1}{2} + \frac{1}{4} = 3/4$ I found the sum of the series so the series is convergent

b) $\displaystyle\sum_{n=1}^{\infty} \frac{5^{n+1}}{4^{2n}}$ $\qquad \dfrac{5^2(5^{n-1})}{16^n}$

$$\sum_{n=1}^{\infty} \frac{25(5^{n-1})}{16(16^{n-1})} \qquad \frac{25}{16}\left(\frac{5}{16}\right)^{n-1}$$

$|r| = \frac{5}{16} < 1$ so convergent

and the sum $= \dfrac{a}{1-r}$

$$S = \frac{25/16}{1-5/16} = \frac{25}{16-5}$$

$25/11$

$b_n = \frac{1}{\sqrt{n^2+3n+1}}$

Compare to $\sum_{n=1}^{\infty} \frac{1}{n}$ a divergent p-series $p = 1 \leq 1$

6. (10 points) Test the following series for absolute convergence, conditional convergence or divergence. State the test you are using and show all your work.

a) $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n^2+3n+1}}$

Limit Comparison Test

$\lim_{n\to\infty} \left(\frac{1}{n} \cdot \frac{\sqrt{n^2+3n+1}}{1} \right) = 1 > 0$ and finite

So Both are divergent

Now Alternating Series Test

i) $\frac{1}{\sqrt{(n+1)^2+3(n+1)+1}} \leq \frac{1}{\sqrt{n^2+3n+1}}$

ii) $\lim_{n\to\infty} \frac{1}{\sqrt{n^2+3n+1}} = 0$

So $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n^2+3n+1}}$ is Conditionally Convergent

b) $\sum_{n=1}^{\infty} \frac{1}{n^2+3^n}$ Compare to

$\sum_{n=1}^{\infty} \frac{1}{3^n}$ which is a convergent geometric series $r = 1/3 < 1$

Direct Comparison Test

$\frac{1}{n^2+3^n} \leq \frac{1}{3^n}$

$3^n \leq n^2+3^n$

$0 \leq n^2$ true for all $n \geq 1$ so

$\sum_{n=1}^{\infty} \frac{1}{n^2+3^n}$ is Absolutely Convergent

c) $\sum_{n=1}^{\infty} \frac{(3\cdot 7\cdot 11\cdot 15\cdots(4n-1))}{5^n\, n!}$

Ratio Test

$\lim_{n\to\infty} \left| \frac{\cancel{3\cdot 7\cdot 11\cdots(4n-1)}(4n+3)}{5\cdot 5^n\,(n+1)\,\cancel{n!}} \cdot \frac{\cancel{5^n\, n!}}{\cancel{(3\cdot 7\cdot 11\cdots(4n-1))}} \right|$

$\lim_{n\to\infty} \left| \frac{4n+3}{5(n+1)} \right| = 4/5 < 1$ So Absolutely Convergent

7. (10 points) The given curve is rotated about the y-axis. Find the area of the resulting surface. $S = 2\pi \int radius\ (arclength)$ $\quad y^{2/3} + x^{2/3} = 1, \quad 0 \le y \le 1$

$$S = 2\pi \int_0^1 (1-y^{2/3})^{3/2} \sqrt{}$$

$$x = (1-y^{2/3})^{3/2}$$

$$dx/dy = \frac{3}{2}(1-y^{2/3})^{1/2}\left(-\frac{2}{3}y^{-1/3}\right)$$

$$= -\frac{\sqrt{1-y^{2/3}}}{y^{1/3}}$$

$$S = 2\pi \int_0^1 (1-y^{2/3})^{3/2} \sqrt{1 + \frac{1-y^{2/3}}{y^{2/3}}}\, dy$$

$$S = 2\pi \int_0^1 (1-y^{2/3})^{3/2} \sqrt{\frac{y^{2/3} + 1 - y^{2/3}}{y^{2/3}}}\, dy$$

$$(dx/dy)^2 = \frac{1-y^{2/3}}{y^{2/3}}$$

$$2\pi \int_0^1 (1-y^{2/3})^{3/2} \sqrt{\frac{1}{y^{2/3}}}\, dy$$

$$2\pi \int_0^1 (1-y^{2/3})^{3/2} \cdot \frac{1}{y^{1/3}}\, dy$$

$$u = 1 - y^{2/3}$$

$$du = -2/3\, y^{-1/3}\, dy$$

$$-\frac{3}{2} y^{1/3}\, du = dy$$

$$2\pi \int -\frac{3}{2} u^{3/2}\, du$$

$$2\pi \cdot -\frac{3}{2} \cdot \frac{1}{5/2} u^{5/2} = 2\pi\left(\frac{-3}{2}\right)\left(\frac{2}{5}\right)(1-y^{2/3})^{5/2}\Big]_0^1$$

$$2\pi\left(-\frac{3}{5}\right)\left[(1-1)^{5/2} - (1-0)^{5/2}\right]$$

$$-\frac{6\pi}{5}[-1] = 6\pi/5$$